Individual assignment 5

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
```

```r
  sheet = "Aquatic Insects"
)

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
      here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
      sheet = "Water Quality",
      na = c("", "n/a", "N/A", "over", "173+")
)
```

Cleaning

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```r
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
```

```r
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

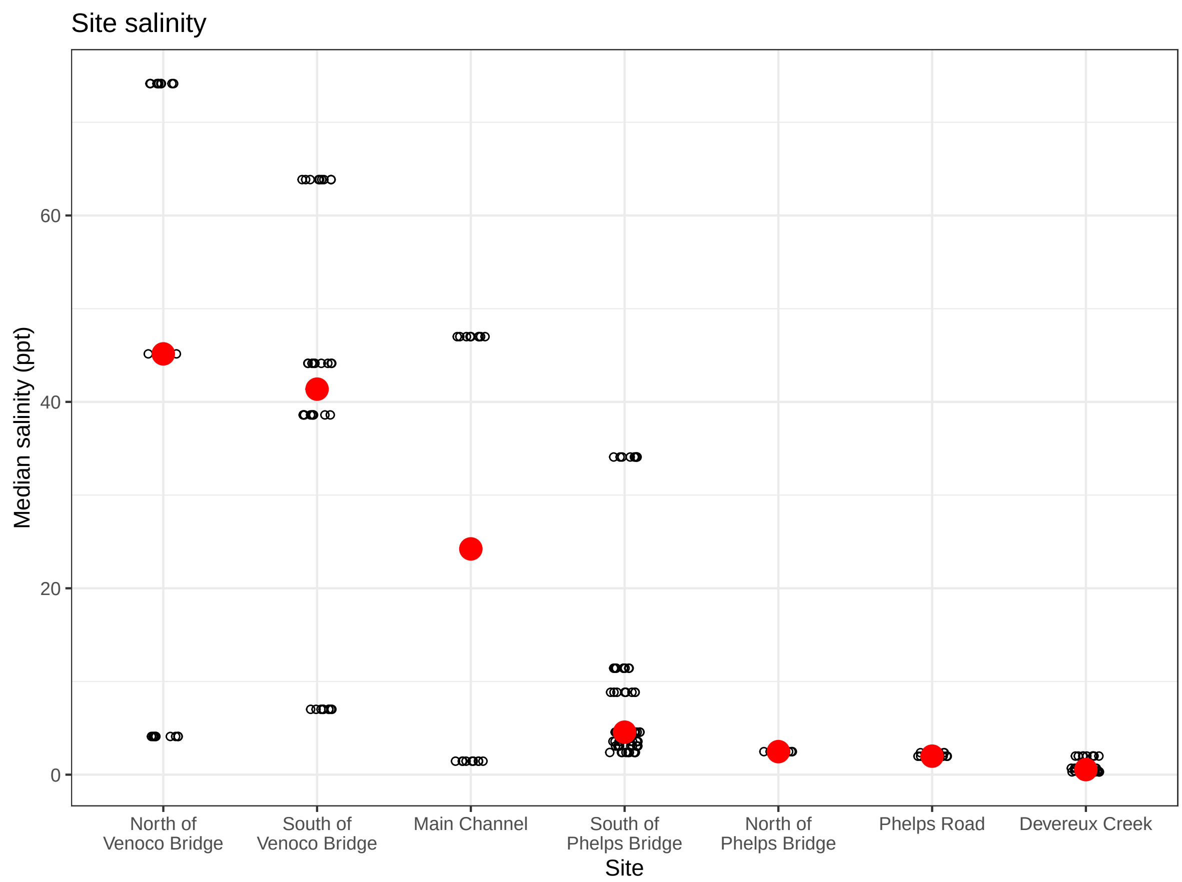
```r
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
```

```r
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(
    site_full, med_sal, .fun = "median", .na_rm = TRUE
    ),
         site_full = fct_rev(site_full))
```

Class visualizations

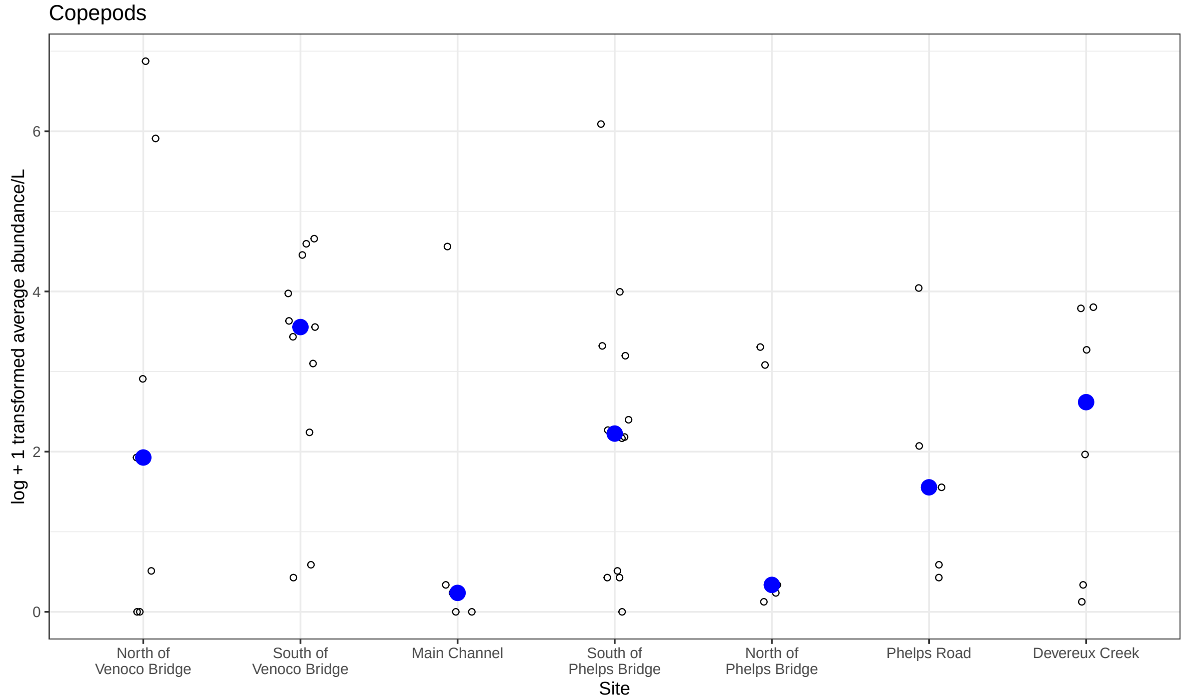
```r
# base layer
salinity <- ggplot(data = aq_ins_clean,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_sal)) +
  # first layer: salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # relabelling x- and y-axes
  labs(x = "Site",
```

```
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  # different theme
  theme_bw()

# displaying object
salinity
```




```
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

```r
# base layer
copepods <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepods") +
  # different theme
  theme_bw()

# displaying object
copepods
```



1. Visualizing differences across sites in major taxon abundance

a.

- x-axis: site
- y-axis: log + 1 transformed average ostracod abundance/L
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()` with a visually distinct point

b.

```
# creating new object from clean aquatic inverts
ostracods <- aq_ins_clean |>
  # filter to only include ostracods
  filter(taxon == "ostracod") |>
  # create log + 1 transformed abundance column
  mutate(log_ave_lit = log(ave_lit + 1))
```

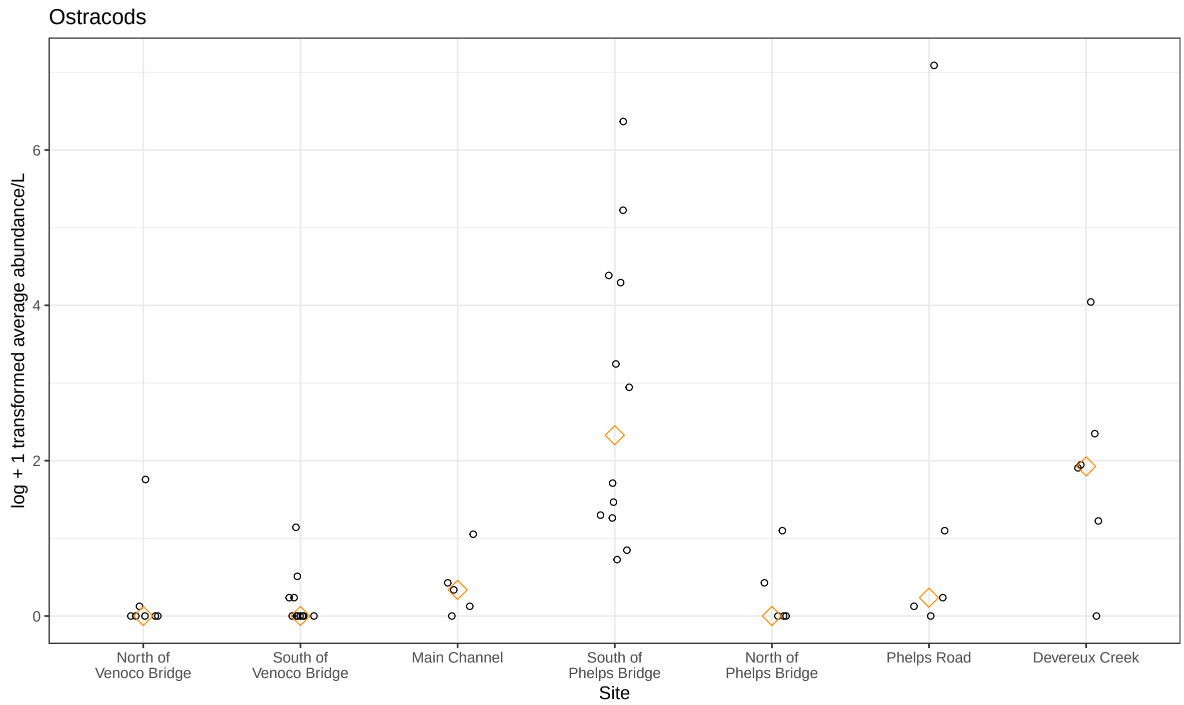
```r
# display 10 random rows from data frame
slice_sample(ostracods, n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2021-11-23_NPB2 2021-11-23 00:00:00 NPB2  ostr~   0.267   7.59    1.99   0.65
 2 2023-08-12_NEC  2023-08-12 00:00:00 NEC   ostr~   0      NA      NA     NA
 3 2021-08-25_NPB  2021-08-25 00:00:00 NPB   ostr~ 581.      7.52    4.57   4.5
 4 2022-05-05_NEC  2022-05-05 00:00:00 NEC   ostr~   0.267   9.45   63.8    7.3
 5 2021-11-16_NPB  2021-11-16 00:00:00 NPB   ostr~  72.1     7.3     2.38  12.1
 6 2022-11-19_NDC  2022-11-19 00:00:00 NDC   ostr~   6      NA      NA     NA
 7 2022-08-12_NEC  2022-08-12 00:00:00 NEC   ostr~   0       7.37   NA     NA
 8 2023-02-28_NPB1 2023-02-28 00:00:00 NPB1  ostr~   0      NA      NA     NA
 9 2023-11-12_NMC  2023-11-12 00:00:00 NMC   ostr~   0.533  NA      47     14.0
10 2020-02-21_NVBR 2020-02-21 00:00:00 NVBR  ostr~   4.8     7.34    4.1    6.1
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c.

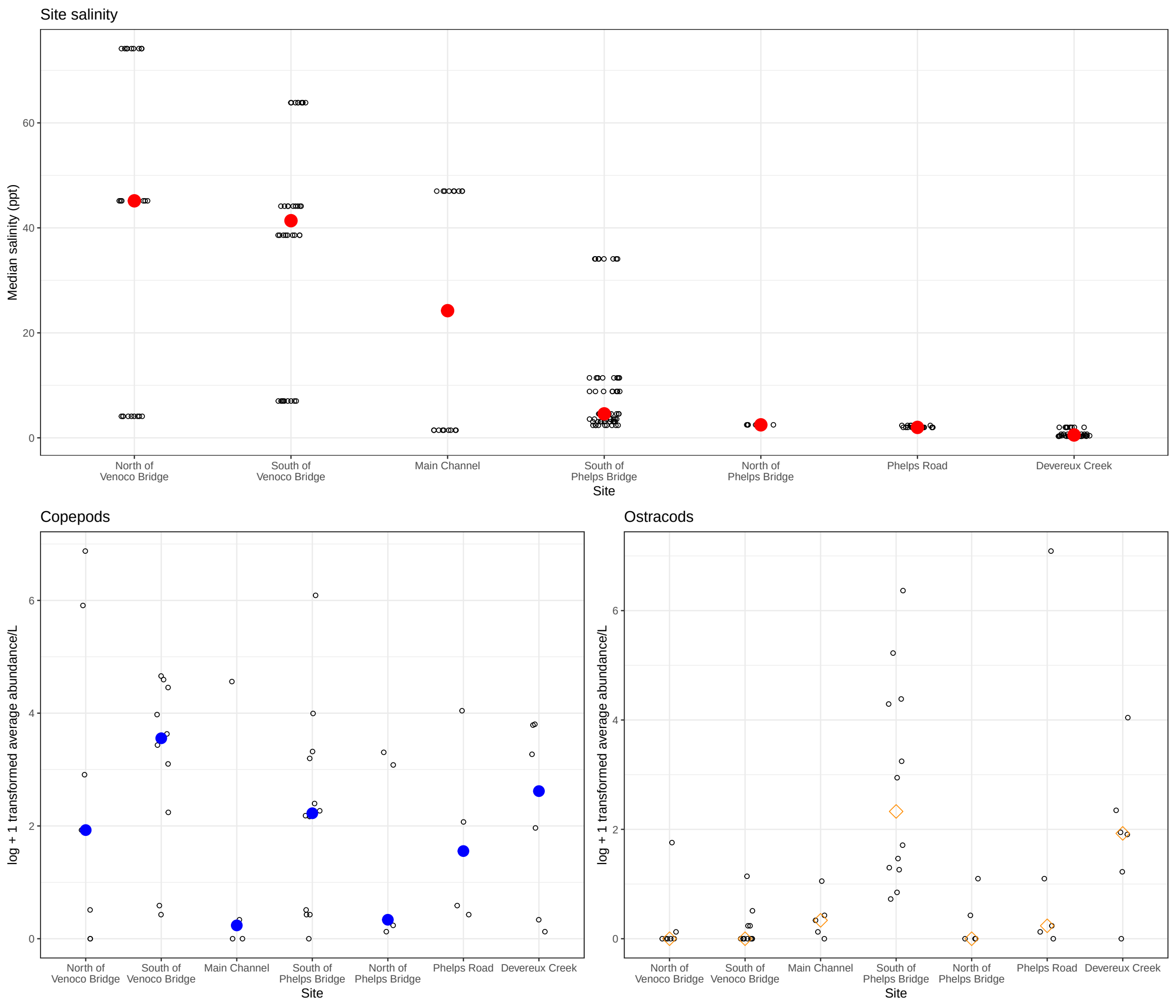
```r
# base layer
ostracods_plot <- ggplot(data = ostracods,
                         # x-axis: site
                         # y-axis: log transformed average abundance/L
                         mapping = aes(x = site_full,
                                       y = log_ave_lit)) +
  # first layer: points to represent individual observations
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # second layer: larger colored diamonds to represent site medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               shape = 23,
               color = "darkorange") +
  # wrap long site names so x-axis labels are readable
```

```
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # relabel axes and add a descriptive title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  # use a non-default theme
  theme_bw()

# display plot
ostracods_plot
```



d.

```
# combine salinity, copepod, and ostracod plots
salinity / (copepods | ostracods_plot)
```



e.

Median copepod abundance is highest at South of Venoco Bridge and Devereux Creek, while median ostracod abundance is highest at South of Phelps Bridge and Devereux Creek. This is shown by the larger colored median points in each panel, as well as the spread of individual observations around those medians.

Ostracod abundance does not appear to closely follow the salinity gradient across sites. For example, South of Phelps Bridge has relatively low median salinity in the top panel but the highest median ostracod abundance in the bottom-right panel, suggesting ostracods may be less strongly associated with salinity than copepods.

2. Vizualizing total abundance as a function of salinity

a.

- x-axis: median salinity, `med_sal`
- y-axis: log + 1 transformed average abundance/L, `log_ave_lit`
- color: taxon
- geometry: `geom_point()`
- facets: `facet_wrap(~ taxon, scales = "free")`

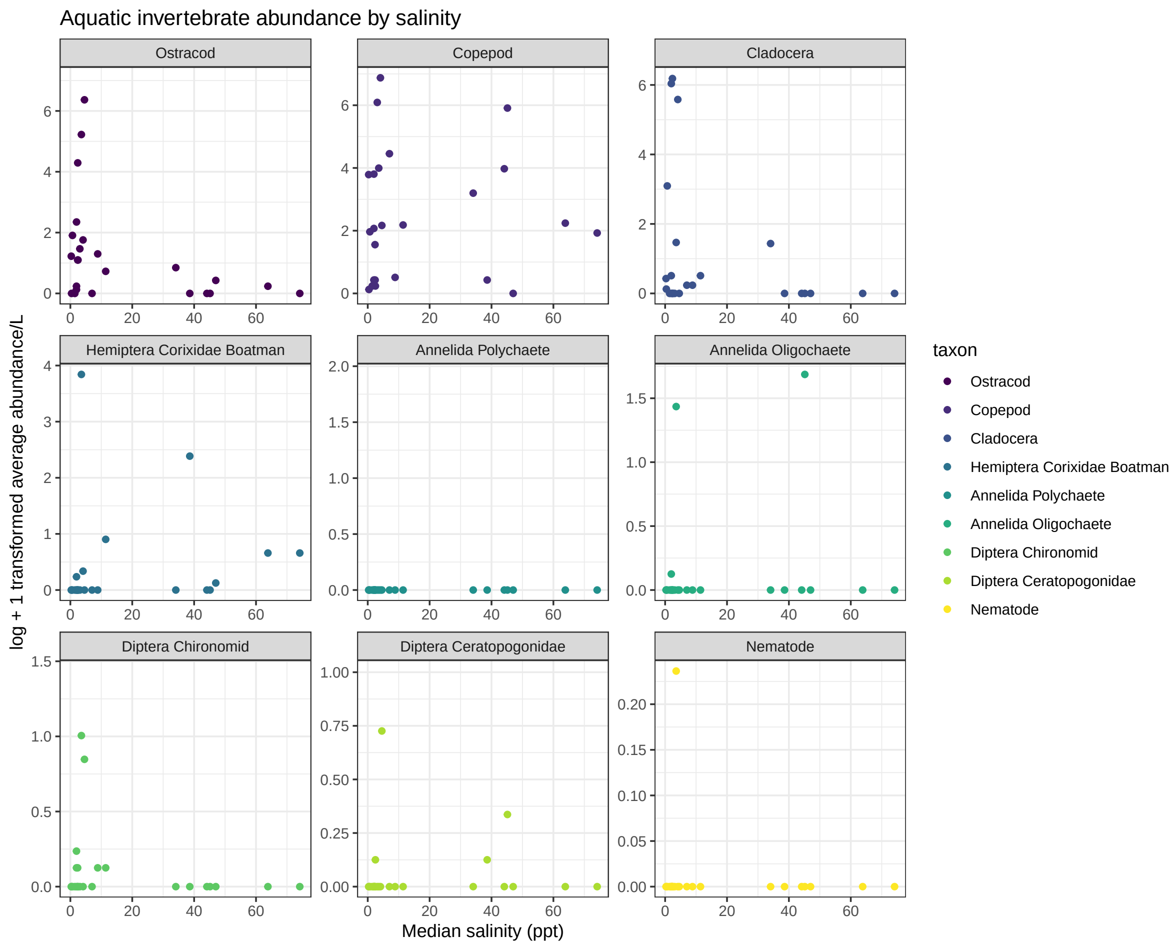
b.

```
# create data frame for abundance-salinity figure
abundance_salinity <- aq_ins_clean |>
  # create log + 1 transformed abundance column
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # change taxon names from snake case to title case
  mutate(taxon = str_to_title(str_replace_all(taxon, "_", " "))) |>
  # order taxon levels by maximum average abundance/L
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
  # keep columns needed for plot
  select(taxon, med_sal, ave_lit, log_ave_lit)

# display 10 random rows from data frame
slice_sample(abundance_salinity, n = 10)
```

```
# A tibble: 10 x 4
   taxon                      med_sal ave_lit log_ave_lit
   <fct>                        <dbl>   <dbl>       <dbl>
 1 Diptera Chironomid            NA     0           0
 2 Diptera Chironomid            NA     0           0
 3 Hemiptera Corixidae Boatman   38.6   9.87        2.39
 4 Diptera Chironomid            11.4   0.133       0.125
 5 Hemiptera Corixidae Boatman   44.1   0           0
 6 Diptera Chironomid            NA     1.33        0.847
 7 Ostracod                      NA     6           1.95
 8 Nematode                      NA     0           0
 9 Hemiptera Corixidae Boatman   11.4   1.47        0.903
10 Diptera Ceratopogonidae       11.4   0           0
```

c

```r
# base layer
ggplot(data = abundance_salinity,
       # x-axis: median salinity
       # y-axis: log transformed average abundance/L
       # color: taxon
       mapping = aes(x = med_sal,
                     y = log_ave_lit,
                     color = taxon)) +
  # points show each taxon's abundance at each salinity value
  geom_point() +
  # separate each taxon into its own panel with free axes
  facet_wrap(~ taxon,
             scales = "free") +
  # use custom colors instead of ggplot defaults
  scale_color_viridis_d() +
  # relabel axes and add title
  labs(x = "Median salinity (ppt)",
       y = "log + 1 transformed average abundance/L",
       title = "Aquatic invertebrate abundance by salinity") +
  # remove legend because taxa are already labeled by facet titles
  theme(legend.position = "none") +
  # use a non-default theme
  theme_bw()
```



d.

Across taxa, abundance appears to vary more strongly at low salinity than at high salinity. Ostracods, cladocerans, diptera chironomids, and annelida oligochaetes have their highest log-transformed abundance values at low salinity, with abundance generally closer to zero at higher salinity values. Copepods show less of a clear negative relationship because some high abundance values also occur at moderate to higher salinity levels. Several taxa, including nematodes, diptera ceratopogonidae, and annelida polychaetes, have mostly low abundance across the salinity range, making their relationship with salinity harder to interpret.